

William Jia-Ang Ou

PhD candidate

Department of Zoology | Biodiversity Research Centre | University of British Columbia

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Education

PhD candidate in Zoology

2021-present

Department of Zoology, University of British Columbia

- Advisor: Dr. Rachel Germain
- Visiting scholar: Fukami lab, Stanford University

Master of Science in Ecology

2021

Institute of Ecology and Evolutionary Biology, National Taiwan University

- Thesis: *Effects of local and landscape factors on paddy field arthropod diversity: from species to communities*
- Advisor: Dr. Chuan-Kai Ho

Bachelor of Science in Biology

2017

Biology Program, University of British Columbia

Fellowships and Grants

Biodiversity Research: Integrative Training and Education (8000CAD)	2026
Graduate Student Research Dissemination Fund (500CAD)	2025
Go Global International Learning Award (1000CAD)	2023
UBC Four Year Fellowship (18200CAD x4)	2022
NSERC BIOS ² Fellow (10500CAD x3)	2021
International Tuition Award (1066CAD)	2021
President's Academic Excellence Initiative PhD Award (325CAD)	2021
Zoology Graduate Fellowship (16000CAD)	2021
Research abroad travel grant (7000NTD)	2019
Faculty of Science International Student Scholarship (5000CAD)	2017
Go Global International Learning Award (2000CAD)	2016

Research experience

Working group co-lead

2021

University of British Columbia, Vancouver, BC

Advisor: Drs. Rachel Germain, Tess Grainger, Dept. Zoology and Biodiversity Research Centre

- Developed a guideline for writing accessible theoretical papers based on cognitive load theory

Research assistant

2020

National Taiwan University, Taipei, Taiwan

Advisor: Dr. David Zeleny, Institute of Ecology and Evolutionary Biology

- Developed a software for quantifying species niche width from co-occurrence data

Research Assistant

2017 - 2020

National Taiwan University, Taipei, Taiwan

Advisor: Dr. Chuan-Kai Ho, Institute of Ecology and Evolutionary Biology

- Examined arthropod trophic interactions using stable isotopes

Visiting Researcher

2019

Kanazawa University, Kanazawa, Japan

Advisor: Dr. Nisikawa Usio, Institute of Nature and Environmental Technology

- Tested biogeographical hypotheses of biodiversity in agroecosystems

Undergraduate Assistant

2016 - 2017

University of British Columbia, Vancouver, Canada

Advisors: Drs. Jennifer Sunday, Joey Bernhardt, Mary O'Connor, Dept. Zoology and Biodiversity Research Centre

- Examined the temperature dependence of resource competition

Directed Studies Student

2016

University of British Columbia, Vancouver, Canada

Advisors: Drs. Daniel Karp, Alejandra Echeverri, Kai Chan, Institute for Resources, Environment and Sustainability

- Quantified the impacts of land use change on avian populations

Publications

* - equal contribution † - shared senior authorship

Preprints

Ou, J-A, Huang, C-L, Chen, H-J, Tsai, C-W, Ho, C-K. 2024. Effects of local and landscape factors on paddy field arthropod diversity: from species to communities. *bioRxiv* [link]

Submitted

Ou, WJ-A, Laver, C., Perez, S., Germain, R. 2026. Ecological memory generates transient dynamics that bias empirical estimates of demographic parameters. *The American Naturalist*

Ou, WJ-A, †Germain, R., †Fukami, T. 2026. Transient species can increase resident metapopulation size by modifying local conditions and regional connectivity. Revisions submitted to *The American Naturalist* (preprint: doi.org/10.64898/2026.01.07.698263)

Ou, WJ-A, Germain, R. 202X. Assortative dispersal facilitates the regional maintenance of alternative stable states. In revision at *Ecological Monographs*

Published

Ou, W. 2026. Bob Holt: "Tools of clear thought". *Population Biology Modeling & Theory* (In press)

Hsu, G-C, **Ou, J-A**, Ni, Min-Hsuan, Lin, Zheng-Hong, Ho, C-K. 2025. Generalist predators function as pest specialists: examining diet composition of spiders and ladybeetles across rice crop stages. *Journal of Applied Ecology* [link]

***Ou, WJ-A**, *Henriques, GJB, Senthilnathan, A, Ke, P-J, †Grainger, T, †Germain, R. 2022. Writing Accessible Theory in Ecology and Evolution: Insights from Cognitive Load Theory. *BioScience*. [link] (Faculty Opinions recommended)

*Lin, P-A, *Liu, C-M, ***Ou, J-A**, Sun, C.-H., Chuang, W.-P., Ho, C.-K., Kinoshita, N., & Felton, G. W. 2021. Changes in arthropod community but not plant quality benefit a specialist herbivore on plants under reduced water availability. *Oecologia*. [link]

Hsu, G-C, **Ou, J-A**, Ho, C-K. 2021. Pest consumption by generalist arthropod predators increases with crop stage in both organic and conventional farms. *Ecosphere* [link]

Invited seminars

Ou, W. 2025 Prospects and limits of computational thinking in biology. CSEE. Sherbrooke, BC

Ou, W. 2023 Ecological theory in service of biological control. Stanford EcoEvo Seminar. Palo Alto, CA

Ou, W. 2022 How to use ecological theory to inform biological control. ESA/ESC/ESBC Joint Annual Meeting. Vancouver, BC

Ou, W. 2022 Computing biology or Biology computing? UBC BLISS seminar. Vancouver, BC.

Conference presentations

Ou, W., Germain, R., Fukami, T. 2026 Transient invaders modify the metapopulation size of a resident species. CSEE. Toronto, ON

Ou, W. 2025 Assortative dispersal facilitates the maintenance of alternative stable states. NetSci. Maastricht, the Netherlands (*Oral*)

Ou, W. 2022 Active aggregation stabilises priority effects and promotes regional coexistence. ESA/CSEE Annual Meeting. Montreal, QC (*Oral*)

Ou, W. 2021 Changes in arthropod community but not resource quality benefits a specialist herbivore under plant water stress. BotZoo Spring Symposium. Vancouver, BC (*Oral*)

Ou, J-A, Huang, C-L, Chen, H-J, Tsai, C-W, Ho, C-K. 2020 Multi-scale analysis reveals the importance of landscape for arthropod biodiversity in a rice agro-ecosystem. Taiwan Entomological Society Annual Meeting. Taipei, Taiwan. (*Oral & Poster*)

Ou, J-A, Huang, C-L, Chen, H-J, Tsai, C-W, Ho, C-K. 2019 Spatial and temporal partitioning of arthropod beta-diversity in rice agro-ecosystems. NTU-NTNU Joint Symposium for Ecology and Evolutionary Biology. Taipei, Taiwan. (*Oral & Poster*)

Ou, J-A, Lin, P-A, Liu, C-M, Sun, C-H, Kinoshita, N., Ho, C-K, Felton, G. W. 2019 Plant drought stress randomizes arthropod community assembly. Congress for Animal Behavior and Ecology. Taipei, Taiwan. (*Oral*)

Ou, J-A 2019 Taxonomic and functional insect herbivore diversity across latitude in rice agro-ecosystems. Congress for Animal Behavior and Ecology. Taipei, Taiwan. (*Poster*)

Ou, J-A, Hsu, G-C, Ho, C-K. 2018 Drivers of arthropod food web dynamics in rice paddy agroecosystem - a stable isotope approach. Ecological Society of Japan. Sapporo, Japan. (*Oral*)

Ou, W. 2017. A test for the habitat-filtering hypothesis: Modeling avian population responses to land-use change in Guanacaste, Costa Rica. Multidisciplinary Undergraduate Research Conference. Vancouver, Canada. (*Oral*)

Teaching and Mentoring

Teaching Assistant

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| • BIOL 336: Fundamentals of Evolutionary Biology, UBC | 2026 |
| • ISCI 320: Research Development Retreat, UBC | 2023-2025 |
| • BIOL 306: Advanced Ecology, UBC | 2021-2025 |
| • LS 5068: General Zoology Laboratory, NTU | 2019 |
| • EEB 5083: Numerical Methods in Community Ecology, NTU | 2019 |
| • EEB 5082: Introduction to R for Ecologists, NTU | 2018 |
| • LS 3023: Community Ecology, NTU | 2018 |

Guest Lecturer

- BIOL 404: Ecological Methodology, UBC 2022–2023
- LS 3023: Community Ecology, NTU 2018

Workshops & Training

- OpenRefine Workshop, Data Carpentry 2024

Undergraduate Mentorship

- Catherine Laver, Honour's student, UBC
- Seth Perez, Work-learn student, UBC
- Sam Deng, Undergraduate Research Experience, UBC
- Ruhi Mander, Undergraduate Research Experience, UBC
- Gen-Chang Hsu, Honour's student, NTU
- Reona Tricia, Work-learn student, NTU

Work experience

BRITE Intern, ESSA Technologies 2026

Data Intern, Living Data Project 2024

Resilience Engineering Intern, VERICA 2022

Research scientist, Melio Inc. 2020

Circular Economy Intern, REnato Lab 2018

Outreach and Service

‡ - rated "best" reviewer

Journal reviewer

- American Naturalist‡, Ecology, Ecology Letters, Evolutionary Applications, New Phytologist, Method in Ecology and Evolution, Proceedings of the Royal Society B, Oikos

Podcast guest, What is Science? Stated casually 2022

Communication committee, BIOS2 Fellowship Program 2022

- Blog post: Computing biology or biology computing?

Invited speaker, National Taiwan Science Education Center 2019

- Talk title: *Biodiversity beyond taxonomic nomenclature* (link; 50:55-1:15)

Editor, Green News Taiwan 2018-2019

Founder, NTU Biodiversity Discussion Group 2018-2020

Teaching Assistant, Biodiversity summer camp, Kinmen National Park 2017

Languages

- Formal: R, Julia, Python (working knowledge)
- Natural: English (native), Mandarin Chinese (native), Taiwanese (proficient)

Software

- PalCreator - an R package for creating your own color palettes [link]
- GeoRef - an R package for georeferencing biodiversity datasets [link]

References

Dr. Rachel Germain

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Biodiversity Research Centre
4200-6270 University Boulevard
Vancouver, BC, V6T 1Z4, Canada
Email: rgermain@zoology.ubc.ca

Dr. Tadashi Fukami

Professor, Departments of Biology and Earth System Science
Stanford University
371 Jane Stanford Way
Stanford, CA, 94305-5020, USA
Email: fukamit@stanford.edu